

Операционные системы

Поиск файлов. Перенаправление ввода-вывода. Просмотр запущенных процессов

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить перенаправление ввода-вывода
- 2 Изучить работу фильтров
- 3 Изучить команду поиска
- 4 Ознакомиться с управлением процессами
- 5 Ознакомиться с командами `df` `du`

2. Процесс выполнения лабораторной работы

```
vmsavostin@vmsavostin:~$ ls /etc/ > file.txt
vmsavostin@vmsavostin:~$ ls >> file.txt
vmsavostin@vmsavostin:~$ cat file.txt
abrt
adjtime
aliases
alsa
alternatives
anaconda
anacrontab
anthy-unicode.conf
asound.conf
at.deny
audit
authselect
avahi
bash_completion.d
bashrc
bindresvport.blacklist
binfmt.d
bluetooth
brlapi.key
brltty
brltty.conf
ceph
chromium
chrony.conf
cifs-utils
```

Рисунок 1: Запись в файл

```
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ grep .conf file.txt > conf.txt  
vmsavostin@vmsavostin:~$ cat conf.txt  
anthy-unicode.conf  
asound.conf  
brltty.conf  
chrony.conf  
cpupower-service.conf  
dconf  
dnsmasq.conf  
dracut.conf  
dracut.conf.d  
fprind.conf  
fuse.conf  
host.conf  
idmapd.conf  
kdump.conf  
krb5.conf  
krb5.conf.d  
ld.so.conf  
ld.so.conf.d  
libaudit.conf
```

Рисунок 2: Поиск расширения .conf


```
/home/vmsavostin/.npm/_cacache/index-v5/c4
/home/vmsavostin/.npm/_cacache/index-v5/1d/f1/ce5e4168f5a5f662c23cbbd558c5d82ef5e57d137e2d329fbc17126408620
/home/vmsavostin/.npm/_cacache/index-v5/68/ca
/home/vmsavostin/.npm/_cacache/index-v5/9e/e0/c3386deef310820ca85762b73df78a4cca889774c30364b036d1f870889
/home/vmsavostin/.npm/_cacache/index-v5/c7
/home/vmsavostin/.npm/_cacache/index-v5/c7/b3/c81fcd46b7c6e984952f72b5d5823025eb825b31eb9b584107047503dd5a
/home/vmsavostin/.npm/_cacache/index-v5/f3/6c/c3986891a2eed646eaa7d914d91cfffcd8d6421272c959bdb5a87486645945
/home/vmsavostin/.npm/_cacache/index-v5/84/ce
/home/vmsavostin/.npm/_cacache/index-v5/09/c0
/home/vmsavostin/.npm/_cacache/index-v5/09/c0/cce94e2c664a8f390a8f1ade3346d5c95f3ced5cf6fc29def30220242c53
/home/vmsavostin/.npm/_cacache/index-v5/ce
/home/vmsavostin/.npm/_cacache/index-v5/59/e7/cb225f160e685933e8439d3cf624ef721a58996cfad55fb2000f59e2edcc
/home/vmsavostin/.npm/_cacache/index-v5/cd
/home/vmsavostin/.npm/_cacache/index-v5/c9
/home/vmsavostin/.npm/_cacache/index-v5/c0
/home/vmsavostin/.npm/_cacache/index-v5/44/44/c9a2c5dc2999cbe868c370ef479e17dfd1e86117d1350482a1d990504757
/home/vmsavostin/.npm/_cacache/index-v5/9f/ae/c077854801f7b4b5cf7eea2bb68ca2bc53f0b53cea310cda0b203217f693
/home/vmsavostin/.npm/_cacache/index-v5/af/cf
/home/vmsavostin/.npm/_cacache/index-v5/c3
/home/vmsavostin/.npm/_cacache/index-v5/a1/5e/c9aa1c46bf0c67511379e97bd07fca87532422f2506214f94db8861160d3
/home/vmsavostin/.npm/_cacache/index-v5/78/34/c1e131014ad3a4ee13f096e70f74c03e68837bc2c8188a44d554f3f3ba2e
/home/vmsavostin/.npm/_cacache/index-v5/ca
/home/vmsavostin/snap/hugo/common
/home/vmsavostin/snap/hugo/current
/home/vmsavostin/conf.txt
vmsavostin@vmsavostin:~$
```

Рисунок 3: Поиск файлов

```
find: '/etc/lvm/archive': Permission denied
find: '/etc/lvm/backup'/etc/foomatic/hashes.d
/etc/foomatic/hashes.d/hashes.new
/etc/hp
/etc/hp/hplip.conf
/etc/httpd
/etc/httpd/conf/httpd.conf
/etc/libibverbs.d/hfi1verbs.driver
/etc/libibverbs.d/hns.driver
: Permission denied
/etc/logrotate.d/httpd
find: '/etc/lvm/cache': Permission denied
find: '/etc/lvm/devices': Permission denied
find: '/etc/nftables': Permission denied
find: '/etc/openvpn/client': Permission denied
/etc/nvme/hostnqn
/etc/nvme/hostid
find: '/etc/openvpn/server': Permission denied
find: '/etc/pki/rsyslog': Permission denied
find: '/etc/polkit-1/localauthority': Permission denied
find: '/etc/polkit-1/rules.d': Permission denied
find: '/etc/sos/cleaner': Permission denied
/etc/sane.d/dll.d/hpaio
/etc/sane.d/hp.conf
/etc/sane.d/hp3900.conf
/etc/sane.d/hp4200.conf
/etc/sane.d/hp5400.conf
/etc/sane.d/hpsj5s.conf
/etc/sane.d/hs2p.conf
:find: '/etc/ssh/sshd_config.d': Permission denied
find: '/etc/sss.d': Permission denied
find: '/etc/sudoers.d': Permission denied
find: '/etc/userdb': Permission denied
```

Рисунок 4: Поиск файлов

```
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ find /etc -name "h*" -print | less  
find: '/etc/audit': Permission denied  
find: '/etc/credstore.encrypted': Permission denied  
find: '/etc/credstore': Permission denied  
find: vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ find ~ -name "log*" > logfile &  
[1] 7626  
vmsavostin@vmsavostin:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
vmsavostin@vmsavostin:~$ rm logfile  
vmsavostin@vmsavostin:~$
```

Рисунок 5: Фоновый запуск процесса

```
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ find ~ -name "log*" > logfile &  
[1] 7626  
vmsavostin@vmsavostin:~$  
[1]+  Done                  find ~ -name "log*" > logfile  
vmsavostin@vmsavostin:~$ rm logfile  
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ gedit &  
[1] 7667  
vmsavostin@vmsavostin:~$ ps | grep gedit  
    7667 pts/0    00:00:00 gedit  
vmsavostin@vmsavostin:~$ kill 7667  
[1]+  Terminated          gedit  
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$
```

Рисунок 6: Фоновый запуск и завершение процесса

```
DF(1)                                User Commands                                DF(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file
    system containing each file name argument.  If no file name is given, the space available on all cur-
    rently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment
    variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the
    space available on that file system rather than on the file system containing the device node.  This
    version of df cannot show the space available on unmounted file systems, because on most kinds of sys-
    tems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes;
        see SIZE format below

    --direct
        show statistics for a file instead of mount point

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 7: Справка по команде df

```
df(1)                                User Commands                                df(1)

NAME
    df - report file system space usage

SYNOPSIS
    df [OPTION]... [FILE]...

DESCRIPTION
    This manual page documents the GNU version of df.  df displays the amount of space available on the file system containing each file name argument.  If no file name is given, the space available on all currently mounted file systems is shown.  Space is shown in 1K blocks by default, unless the environment variable POSIXLY_CORRECT is set, in which case 512-byte blocks are used.

    If an argument is the absolute file name of a device node containing a mounted file system, df shows the space available on that file system rather than on the file system containing the device node.  This version of df cannot show the space available on unmounted file systems, because on most kinds of systems doing so requires non-portable intimate knowledge of file system structures.

OPTIONS
    Show information about the file system on which each FILE resides, or all file systems by default.

    Mandatory arguments to long options are mandatory for short options too.

    -a, --all
        include pseudo, duplicate, inaccessible file systems

    -B, --block-size=SIZE
        scale sizes by SIZE before printing them; e.g., '-BM' prints sizes in units of 1,048,576 bytes; see SIZE format below

    --direct
        show statistics for a file instead of mount point

Manual page df(1) line 1 (press h for help or q to quit)
```

Рисунок 8: Запуск команды df

```
vmsavostin@vmsavostin:~$  
vmsavostin@vmsavostin:~$ df  
Filesystem      1K-blocks    Used Available Use% Mounted on  
/dev/nvme0n1p3 155186176 91465840 63132800 60% /  
devtmpfs        4011280      0 4011280  0% /dev  
tmpfs           4035656      92 4035564  1% /dev/shm  
tmpfs           1614264     1944 1612320  1% /run  
tmpfs            1024         0 1024  0% /run/credentials/systemd-journald.service  
tmpfs           4035656     1640 4034016  1% /tmp  
/dev/nvme0n1p3 155186176 91465840 63132800 60% /home  
/dev/nvme0n1p2 1992552     518800 1352512  28% /boot  
/dev/loop1      115712     115712      0 100% /var/lib/snapd/snap/hugo/25676  
/dev/loop2       49280      49280      0 100% /var/lib/snapd/snap/snapd/25935  
/dev/loop0       68480      68480      0 100% /var/lib/snapd/snap/core24/1349  
tmpfs            1024         0 1024  0% /run/credentials/systemd-resolved.service  
tmpfs           807128      168 806960  1% /run/user/1068  
/dev/sr0         2677920    2677920      0 100% /run/media/vmsavostin/Fedora-WS-Live-43  
vmsavostin@vmsavostin:~$
```

Рисунок 9: Справка по команде du

```
4      ./npm/_cacache/index-v5/a7
4      ./npm/_cacache/index-v5/d4/09
4      ./npm/_cacache/index-v5/d4
4      ./npm/_cacache/index-v5/30/1a
4      ./npm/_cacache/index-v5/30
424    ./npm/_cacache/index-v5
115644 ./npm/_cacache
115708 ./npm
12     ./snap/hugo/25676/.config/go/telemetry/local
0      ./snap/hugo/25676/.config/go/telemetry/upload
12     ./snap/hugo/25676/.config/go/telemetry
12     ./snap/hugo/25676/.config/go
12     ./snap/hugo/25676/.config
12     ./snap/hugo/25676
0      ./snap/hugo/common
16     ./snap/hugo
16     ./snap
0      ./monthly
0      ./reports/monthly
0      ./reports
4      ./ski.plases/equipment
0      ./ski.plases/plans
4      ./ski.plases
0      ./australia
0      ./play/games/play
0      ./play/games
0      ./play
1444872 .
vmsavostin@vmsavostin:~$
```

Рисунок 10: Запуск команды du


```
/home/vmsavostin/.npm/_cacache/index-v5/ca
/home/vmsavostin/.npm/_cacache/index-v5/ca/9d
/home/vmsavostin/.npm/_cacache/index-v5/4b
/home/vmsavostin/.npm/_cacache/index-v5/4b/ef
/home/vmsavostin/.npm/_cacache/index-v5/52
/home/vmsavostin/.npm/_cacache/index-v5/52/4d
/home/vmsavostin/.npm/_cacache/index-v5/85
/home/vmsavostin/.npm/_cacache/index-v5/85/e2
/home/vmsavostin/.npm/_cacache/index-v5/a7
/home/vmsavostin/.npm/_cacache/index-v5/a7/81
/home/vmsavostin/.npm/_cacache/index-v5/d4
/home/vmsavostin/.npm/_cacache/index-v5/d4/09
/home/vmsavostin/.npm/_cacache/index-v5/30
/home/vmsavostin/.npm/_cacache/index-v5/30/1a
/home/vmsavostin/snap
/home/vmsavostin/snap/hugo
/home/vmsavostin/snap/hugo/25676
/home/vmsavostin/snap/hugo/25676/.config
/home/vmsavostin/snap/hugo/25676/.config/go
/home/vmsavostin/snap/hugo/25676/.config/go/telemetry
/home/vmsavostin/snap/hugo/25676/.config/go/telemetry/local
/home/vmsavostin/snap/hugo/25676/.config/go/telemetry/upload
/home/vmsavostin/snap/hugo/common
/home/vmsavostin/monthly
/home/vmsavostin/reports
/home/vmsavostin/reports/monthly
/home/vmsavostin/ski.places
/home/vmsavostin/ski.places/equipment
/home/vmsavostin/ski.places/plans
/home/vmsavostin/australia
/home/vmsavostin/play
/home/vmsavostin/play/games
/home/vmsavostin/play/games/play
vmsavostin@vmsavostin:~$
```

Рисунок 11: Поиск директорий

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.